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From Surveillance to Solution: A Unified Approach to Well and Reservoir Integrity in a Mature EOR Asset

Pranay Shrivastava, Manuel Guzman, Junghun Leem, Avinash Bohra, and Ritik Raj, Cairn Oil & Gas, Vedanta Ltd, Barmer, Rajasthan, India.

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Abstract

This paper introduces a comprehensive, field-validated methodology that bridges well integrity and reservoir integrity through a unified, multidisciplinary framework. Designed for mature brownfield operations, the approach redefines the conventional role of the well integrity teams by integrating geoscience, drilling, and production data into a three-tiered structure: monitoring, diagnosis, and corrective action. The framework leverages both conventional indicators—such as annulus pressure, temperature anomalies, and corrosion logs—and unconventional datasets, including caliper logs, induced seismicity, and geomechanical modeling. Reservoir pressure trends from infill drilling and depletion studies further enrich the diagnostic process.

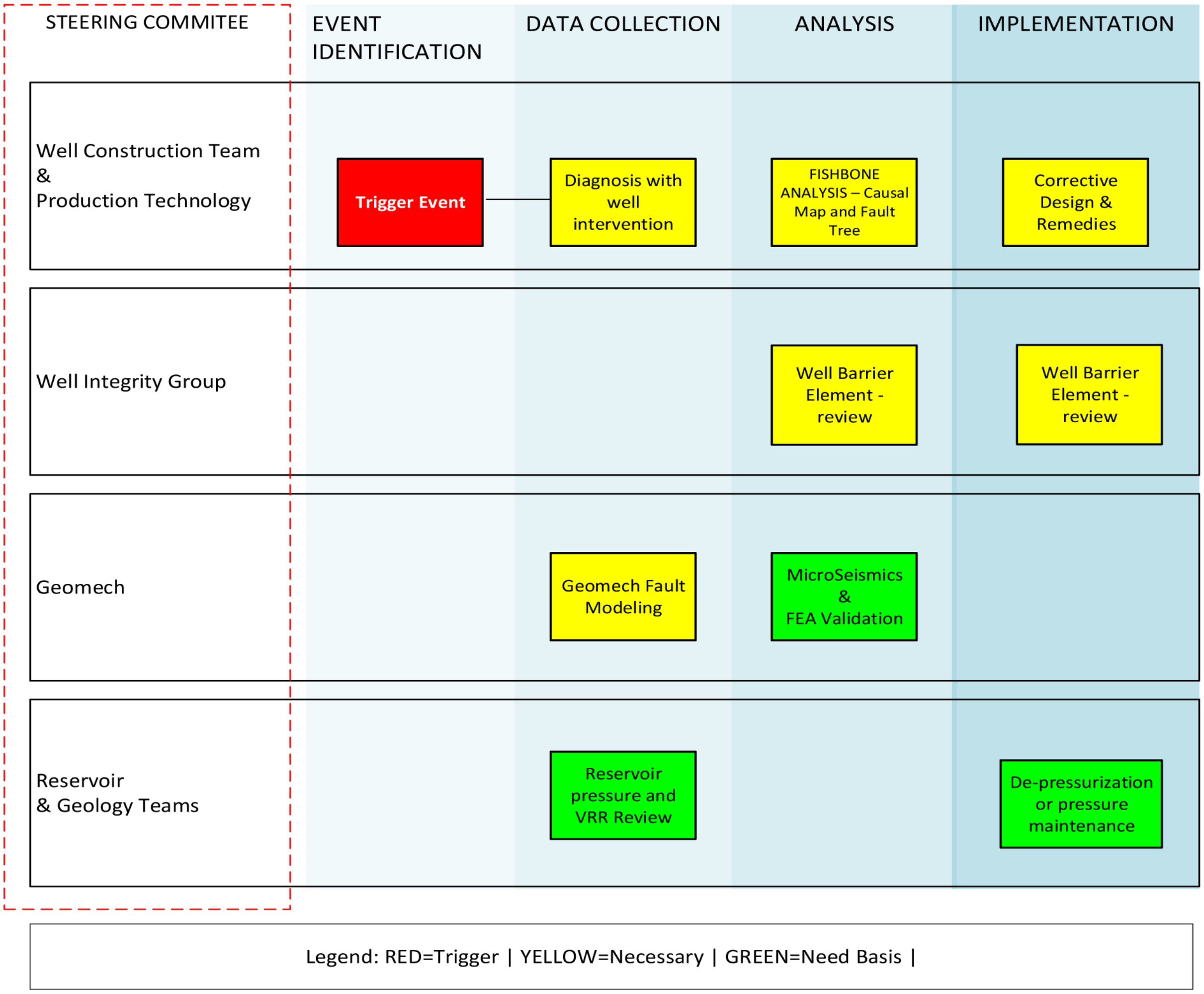
By enabling cross-functional collaboration, the methodology facilitates accurate classification of integrity threats, ranging from corrosion-induced leaks to geomechanical instabilities. Tailored corrective actions—such as remedial cementing, pressure rebalancing, and chemical inhibition—are deployed based on root cause analysis. Field applications demonstrate the framework's adaptability and effectiveness: in one case, point leaks were identified, prompting a revised production strategy. In another, a geo-mechanical failure was correctly identified with multi-disciplinary detailed investigation, avoiding costly sidetracking.

Introduction

The methodology is demonstrated through detailed resolution of integrity issues in two wells—Well #A and Well #B—located in the Mangala field, operated by Cairn Oil & Gas in the RJ-ON 90/1 block, Rajasthan, India. Mangala is one of the largest polymer Enhanced Oil Recovery (EOR) projects globally, reference (Dhruva 2014), characterized by multi-Darcy sandstones with high porosity. The Mangala Field, situated within the Barmer Basin, is a mature yet high-potential hydrocarbon asset discovered in 2004. Characterized by a geologically intricate setting, the field's structural complexity is largely governed by the Main Boundary Fault at the base producing sandstone formation. This fault system plays a critical role in reservoir compartmentalization and significantly influences fluid flow dynamics. The field employs both edge and pattern injection drives for pressure maintenance, with oil producers completed using dual casing designs and primarily lifted via jet pumps. While detailed well designs are beyond the scope of this paper, relevant design elements are discussed as needed to illustrate the framework's application.

Multi-Disciplinary Framework - Detecting, Analyzing & Solving

A multidisciplinary framework developed to streamline the process of managing well integrity in oil and gas field is shown in Figure 1. The framework is anchored by a steering committee comprising four specialized engineering domains: Well Construction & Production Technology, Well Integrity, Geomechanics, and Reservoir & Geology. Each domain plays a distinct and complementary role in resolving complex subsurface challenges. The workflow is structured into four sequential stages. It begins with Event Identification, where anomalies such as annulus pressure or casing deformation are flagged primarily by the Well Construction team or Well integrity team. This is followed by Data Collection, involving diagnostic interventions, barrier element reviews, geomechanical modeling, and reservoir pressure maps. The Analysis phase integrates real-time mapping, microseismic evaluations, and risk assessments to interpret the collected data. Finally, the Implementation stage translates insights into corrective actions such as cement squeezes, barrier reinforcements, and pre-emptive production maintenance. The framework's color-coded legend —*Red* for triggers, *Yellow* for necessary actions, and *Green* for need-based tasks—ensures clarity and prioritization across teams. This integrated approach fosters collaboration, enhances diagnostic precision, and supports proactive decision-making, ultimately improving well integrity and operational efficiency.



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EVENT TRIGGER
1. Abnormal trigger in Annulus pressure
2. Casing deformation
3. ALS failure:
    a. Jet Pump power fluid rate increase
    b. PCP rod torque spike
    c. ESP with packer completion – annulus pressure anomaly

2

DATA ACQUISITION

WELL CONSTRUCTION
1. Diagnostic: Temperature & Noise Logs
2. Annulus and Tubing pressure test.
3. Workover – caliper logs/corrosion logs

WELL INTEGRITY
1. Re-visit Well barrier envelopes
2. Cement log quality check
3. Injection pressures v/s cap rock strength
5. Drilling anomalies in neighboring area

GEOMECH
1. Fault mapping at deformation depth
2. Seismic features
3. Mechanical Earth Model (MEM)
4. Fault critical stress
5. Micro-seismic event in area

RESERVOIR & GEOLOGY
1. Overpressure/under-pressure data
2. Voidage replacement ratios in area

3

ANALYSIS
PRODUCTION TECHNOLOGY
1. Defect elimination – metallurgy & stress check

WELL INTEGRITY
1. Leak or deformation depth
2. Nature of leak – Corrosion or Seismicity
3. Fluid migration through cement channels.

GEOMECH
1. Seismic events
2. Possible fault reactivation

Figure 1—Integrated Workflow for integrity event investigation and responsibilities of steering committee

Demonstration of Integrity Framework in Integrity Resolution

The paper consists of detailed resolution of integrity issues in two different oil producer wells – Well #A & Well #B, in Mangala field. The oil producer wells are completed with two casing design – surface casing and production casing. Majority of oil producer wells operate on Jet Pump as artificial lift system with power fluid (treated produced water) pumped via the annulus. The detailed well design is beyond the scope of this paper but the designs are elaborated on need-basis to demonstrate integrity framework.

CASE-A: Well #A - Investigation using Integrated Workflow

Well #A is an oil producer in Mangala field. It was drilled and completed in 2014, The well has a conductor of 20″, surface casing of 13-3/8″ L80 and production casing of 9-5/8″ 47ppf L80. The well is further completed with 4-1/2″ 12.6 ppf L80 13Cr with a 4-1/2″ × 9-5/8″ production packer. The well is on Jet Pump artificial lift. Power fluid is pumped in reverse circulation from 4-1/2″ × 9-5/8″ annulus through a circulation SSD in production tubing. Jet Pumps are installed in production tubing with throat and nozzle appropriately sized for completion design and productivity index of well. There are 4 dominant zones in field – shallow sands where aquifer is location, claystone formation, tight sandstone formation and main pay zone originated from fluvial deposition environment. The surface casing shoes are situated in clay stones. The production casing is set through tight sand stones and main pay zones.

‘A’ casing pressures and ‘B’ casing pressures are regularly monitored through instrumentation which are relayed to in-house WIMS (Well Integrity Management System) software. A Traffic light based alert system is set on all wells in this field.

- Green - Annulus pressures less than MINAP (Minimum allowable annulus pressure ~ 50 psig for ‘B’ Annulus)
- Yellow – SAP (Sustained Annulus Pressures) between 50 psig and 20% MAASP (Maximum allowable annulus surface pressures)
- Orange – SAP between 20% MAASP to 80% MAASP
- Red – SAP > 80% MAASP

STEP-1 Event Trigger

On Well #A. an alarm on ‘B’ Annulus pressure got triggered, indicating ‘B’ Annulus pressures was above MAASP (Figure 2). Field operations team immediately attempted to bleed and lube as per protocol, and found water in return bleed samples. The laboratory test showed similarity to water being used as power fluid. Salinity and ionic analysis (Ca, Mg, bicarbonates) are excellent markers. When the power fluid was cut off in Annulus ‘A’, both annuli ‘A’ & ‘B’ casing reduced synchronously (Figure 3)

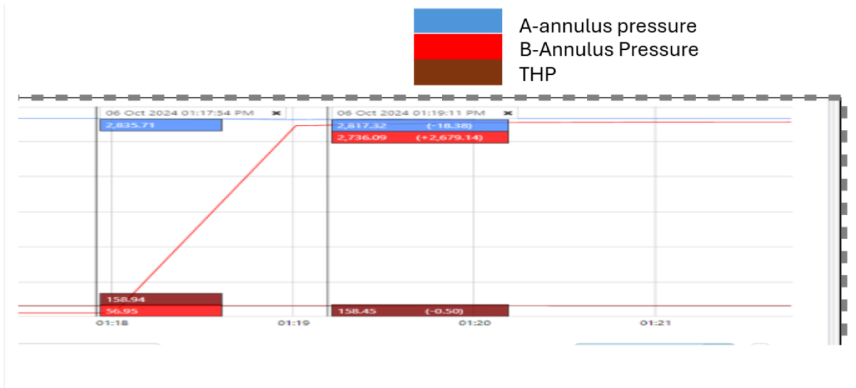

Figure 2—Event trigger - 'B' annulus pressure increased and equalized to 'A' Annulus pressure

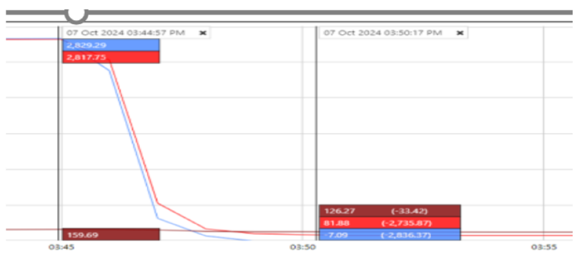

Figure 3—'A' & 'B' Annulus Pressure reduced after well shut down

STEP-2 Data Collection & safe operating envelope review

Well #A was immediately shut down post trigger event. It was now important to initiate collecting data and review the Well #Barrier envelope of the well. The RCA (Root Cause Analysis) team was immediately alerted. Two failure hypotheses were found possible – firstly leak in production casing above surface casing shoe depth or secondly an external force damaging both surface casing and production casing. The following tasks were taken by three teams for investigation by the Well Integrity team to review the well-barrier envelope.

Well Construction Team	Geomechanics team	Reservoir & Geology Team
1. Annulus hold up test 2. Tubing integrity test 3. Tubing-Annulus envelope test 4. Temperature logs 5. Caliper logs 6. Corrosion logs 7. Noise Logs	1. Obtain leak/deformation depth from Well construction team. 2. Obtain nature of damage – a leak, a pin hole or a significant deformation 3. Study seismic features at depth of investigation.	1. Study of over-pressure / depletion based on reservoir surveillance. 2. Study of over-pressure/depletion in overburden sections based on infill drilling campaign in surrounding area – mud logs, kicks, losses, etc.

Briefly above tasks are explained below to understand their relevance.

1. **Annulus hold up test:** The Annulus Hold Test (AHT) is conducted to evaluate the integrity of the annular space, particularly the 'A'-annulus, in oil and gas wells. The jet-pump is retrieved from circulation SSD, SSD is then closed, and production tubing/casing annulus is then tested to observe leak in either in tubing or casing.
2. **Tubing integrity test:** After above test, a lock mandrel with plug is then set in a nipple profile below the production packer in the production tubing. The tubing is then pressure tested.
3. **Tubing-Annulus envelope test**: The plug is set in nipple profile below production packer, in production tubing. The SSD is kept open. After this, the entire envelope of casing and tubing is then tested.

The three tests mentioned above in combined form can establish whether there is a leak in tubing, casing or both. Depth cannot be established, however. Depth sensitive diagnostics tests are then run.

4. **Temperature logs:** Temperature logging, when combined with cold-water pumping into the 'A'-annulus, is an effective diagnostic method to detect and locate casing leaks in oil and gas wells.The fundamental principle relies on thermal contrast: when cold water is injected into the 'A'-annulus, it creates a distinct temperature profile along the wellbore. If a casing leak exists, the injected water escapes into the surrounding formation, leading to a localized cooling effect at the leak point. In case of this investigation, cold water is chosen as jet pumping operates on hot fluid in Mangala field.

   Other methods mentioned in the list of tasks will be elaborated subsequently in this paper in sequence of their application during investigation.

STEP-3 Analysis

WELL CONSTRUCTION GROUP. Tubing integrity test at 2000 psig showed that there was no pressure drop in 15 minutes, confirming the integrity of production tubing. Annulus 'A' was then tested at 500 psig for 15 minutes, indicating a sharp rise in 'B' Annulus pressure.

Well Construction team ran memory-based temperature log inside production tubing before and after cold water pumping in Annulus 'A'. The temperature profile merged at 120-125 mMD indicating that the leak point is at this depth, causing all cold water leaving the well envelope at leak depth. Figure 4

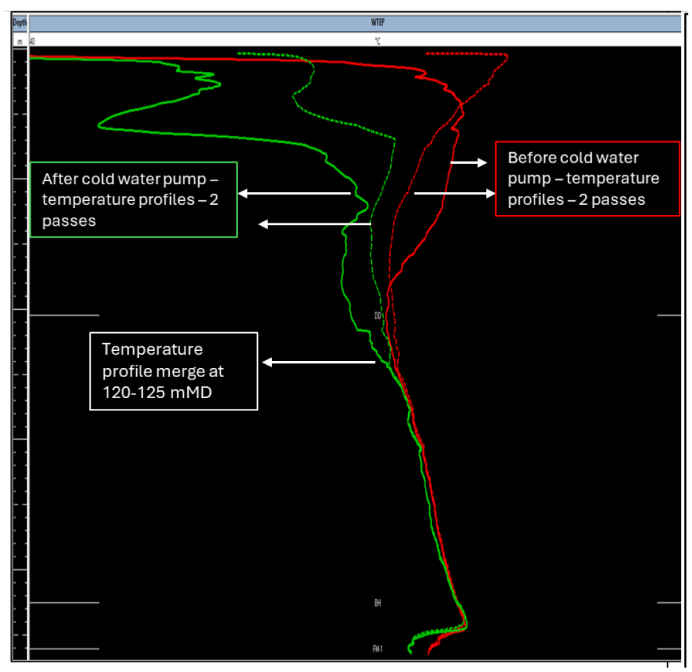

Figure 4—Temperature diagnostic logs in production tubing

A noise log was then run inside production tubing with baseline recorded with no flow for background noise. A subsequent log as run with 'B' Annulus under controlled bleed off to record noise of leak. In noise logs in below figure, a low-frequency leak noise was identified at same depth ~ 120 mMD (Figure 5). Noise logging is a passive acoustic diagnostic technique used to detect and locate leaks or flow paths in oil and gas wells. It relies on capturing low-frequency sound waves generated by fluid or gas movement through restrictions, such as leaks, cracks, or failed seals. Baseline (Road Noise) Run: captures ambient or "road" noise — background sounds from the wellbore, casing, and surrounding formation. The well is then flowed or pressured up to induce movement through any potential leak paths. In figure below, the red-circled area labeled *"Very low intensity/low frequency noise observed at slower speed and stations adjacent to road* *noise"* suggests subtle acoustic anomalies that may indicate minor flow or early leak development. Another red circle highlights *"Clear Noise detected confirming Leak Depth"*, which is the most critical finding. This is where the acoustic signature distinctly deviates from the baseline, confirming active flow through a leak path.

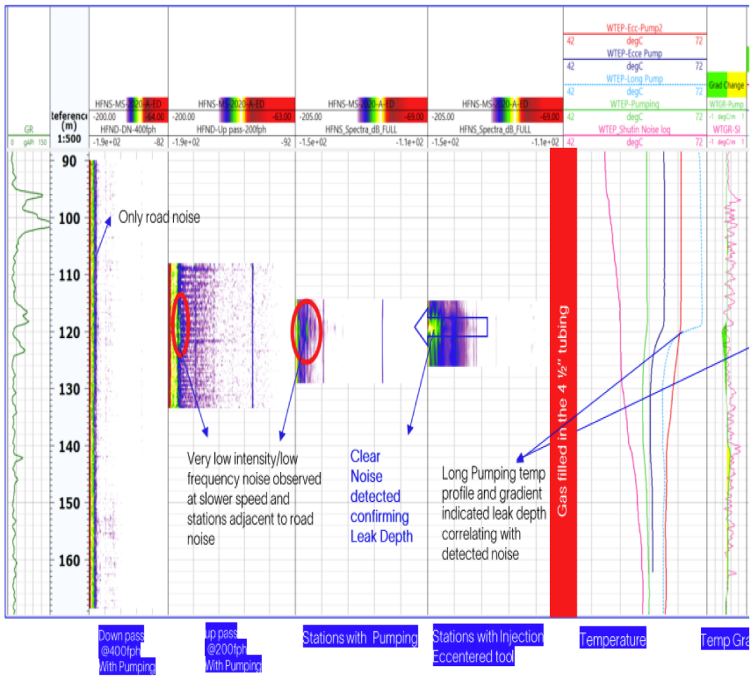

Figure 5—Noise logs run inside production tubing with Annulus B Bleed off catch low-frequency leak noise behind two strings

This raised suspicion on well casing health, hence 4-1/2″ completion was pulled out and an ultra-sonic tool was run on corrosion mode on wireline cable. The findings of corrosion logs Figure 6 suggested metal loss in certain areas. The Geomechanics team was asked not to investigate further on rock deformation hypothesis. Production technology team was alerted to re-run suitability of metallurgy on power fluid used in the well. Figure 6 shows the fishbone approach used, and all possible factors involved in leak of production casing (Figure 7). For the brevity of the paper here, the details of the investigation are not mentioned here. But the team came up with possible microbial corrosion in this specific case. The coupon installed on power fluid lines were also inspected as part of analysis. The corrosion rate was Well #Below design criteria confirming the efficacy of chemical dosing plan in surface lines for surface facility reliability. However, it was decided to repair the Well #And bring it on production with a new completion where annulus is not exposed to energized fluid thus ensuring the longevity of well.

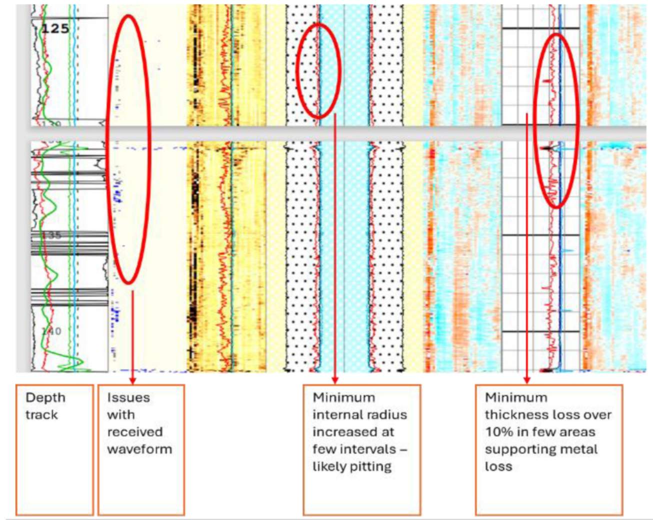

Figure 6—Ultra-Sonic corrosion logs Well#A

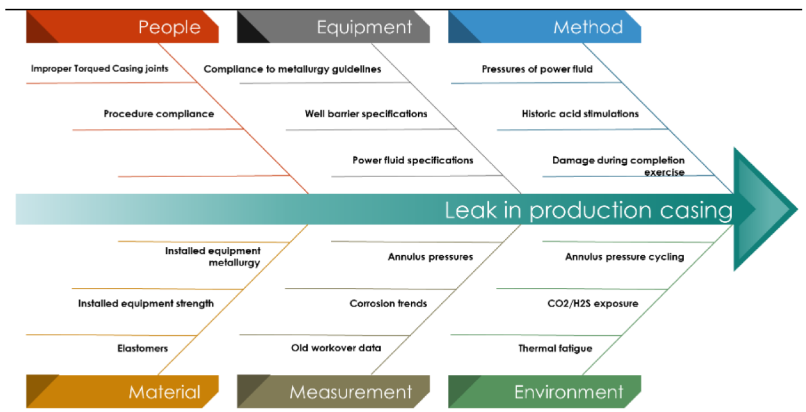

Figure 7—Fishbone approach used to find reasons for leaked casing

WELL INTEGRITY GROUP. Well integrity team suggested to install casing patch in this Well #And changing artificial lift mode to PCP (Progressive Cavity Pump). In such conditions, the annulus will not be affected by fluid, pressure and temperature cycling. The final packer fluid will contain corrosion inhibitors and biocides. Models were re-run to understand the effect on production.

Understanding the status after years of production and/or injection is critical to ensure wells are actively monitored and risk profile is up to date. The Well Barrier envelope is the combination of the individual completion components that ensure all produced fluids are contained and controlled throughout the life of the well. Well Integrity team created Well #Barrier element diagram for both phases to study well integrity status. Non descriptive diagrams are attached below (Figure 8). It is a recommended practice to keep a record of well integrity status of wells through Well #Barrier diagrams. It is company policy to operate all the well with 2 competent barriers all the time in all conditions.

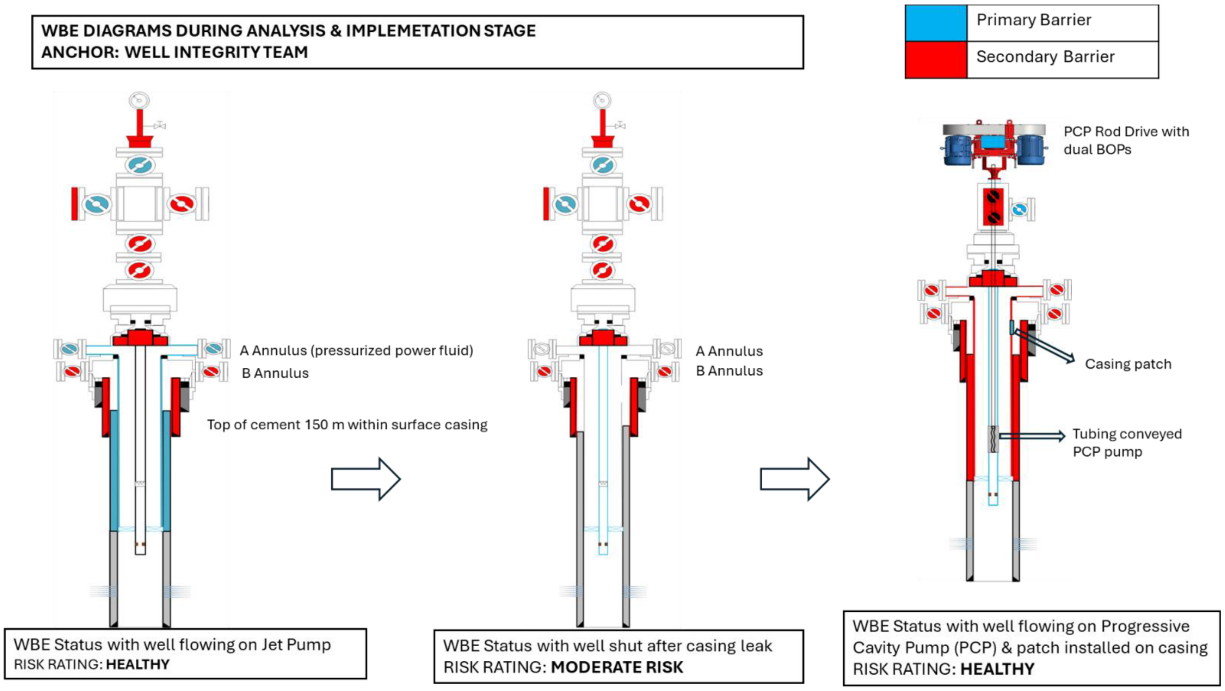

Figure 8—Well Barrier Envelope for Well #A

STEP-4 Implementation

A final re-design of wells was approved and to be implemented based on above workflow.

1. Well recompleted as PCP (Progressive cavity pump)
2. Annulus is no more pressurized.
3. Thin metallic expandable casing patch to be installed inside well.
4. Localized pitting to be investigated in surrounding wells with help of corrosion logs recording during workover.

CASE-B: WELL #B: Investigation using Integrated Workflow

Well #B is an oil producer in Mangala field. It was drilled and completed in 2019, The well has a conductor of 20″, surface casing of 13-3/8″ and production casing of 9-5/8″ 47ppf L80. The well is further completed with 4-1/2″ 12.6 ppf L80 13Cr with a 4-1/2″ × 9-5/8″ production packer. The well is on Jet Pump artificial lift. Power fluid is pumped in reverse circulation from 4-1/2″ × 9-5/8″ annulus through a circulation SSD in production tubing. In short, the well design is similar to that of previous well mentioned in CASE-A.

STEP-1 Event trigger

In 2024, the Well Monitoring team observed that power fluid pressures on ‘A’-Annulus (used for jet pumping operations) is 700 psig below the surrounding wells. While other wells with similar completion and similar productivity were operating on 2900 psig jet pump pressures, Well #B was operating on 2200 psig. Please note that the surface facility injection pressure at header is 2900 psig. ‘B’ Annulus pressures were healthy. This immediately gave RCA team a hypothesis that there was a leak in production casing but below surface casing shoe. As a result, the power fluid is being lost to some other zone behind production casing. The hypothesis was tested with the same workflow.

STEP-2 Data Collection & safe operating envelope review

The well was immediately shut down. It was now important to initiate collecting data and review the Well #Barrier envelope of the well. The RCA team was immediately alerted. Two failure hypotheses were found possible – firstly leak in production casing above surface casing shoe depth or secondly an external force damaging both surface casing and production casing.

The following tasks were taken by three teams for investigation by the Well Integrity team to review the well-barrier envelope for this case.

Well Construction Team	Geomechanics team	Reservoir & Geology Team
1. Annulus hold up test 2. Tubing integrity test 3. Tubing-Annulus envelope test 4. Temperature logs 5. Cement bond logs 6. Multi-arm caliper log	1. Obtain leak/deformation depth from Well construction team. 2. Obtain nature of damage – a leak, a pin hole or a significant deformation 3. Study seismic features at depth of investigation.	1. Study of over-pressure / depletion based on reservoir surveillance. 2. Study of over-pressure/depletion in overburden sections based on infill drilling campaign in surrounding area – mud logs, kicks, losses etc.

The details of these tests are elaborated in CASE-A investigation.

STEP-3 Analysis

WELL CONSTRUCTION TEAM. Tubing integrity test at 2000 psig showed that there was no pressure drop in 15 minutes, confirming the integrity of production tubing. Annulus ‘A’ was then tested at 1500 psig for 15 minutes. Drop in ‘A’ Annulus pressure was observed but the tubing head pressure and ‘B’ Annulus pressure were steady. A leak in production casing below surface shoe and above production packer was confirmed. An envelope test was not required after the above tests.

Next a temperature log was run to investigate the depth of leak. Figure 9 below shows that temperature profile before and after pumping cold water merged at 1181 m MD. Conforming leak depth at 1181 m MD. Next it was decided to understand the nature of damage, hence the Workover team pulled out completion and ran corrosion logs and multi-arm caliper logs. The corrosion logs were healthy ruling out corrosion issue but found shear deformation in production casing. Figure 10 depicts the 24-arm caliper log casing profile around investigation depth. The figure is a casing internal image formed by interpolating casing ID measures with 24 caliper data. The impinged side of casing is shown in blue color while the diametrically opposite site that is bulged out is shown in red color. This image shows a lateral impingement of 0.4 inches.

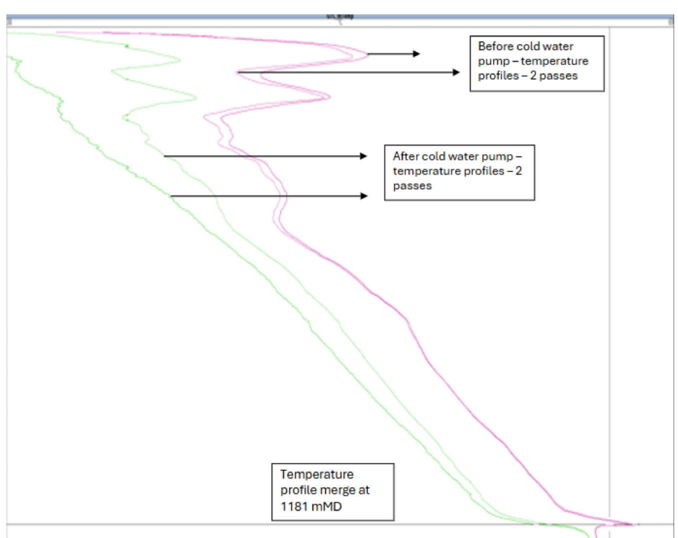

Figure 9—Temperature diagnostic logs in production tubing

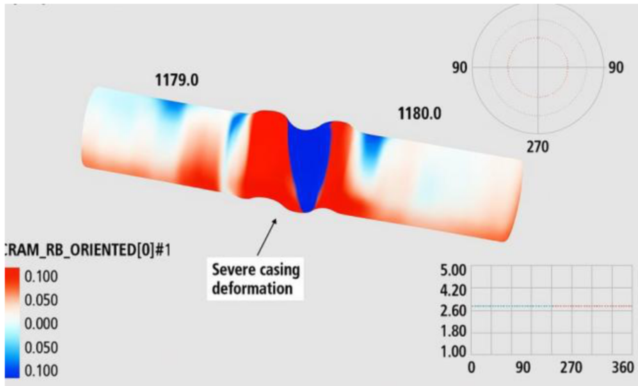

Figure 10—Multi arm caliper casing profiling

Cement bond logs were also evaluated and found that cement around deformed depth is missing. The same is shown in Figure 11. Injectivity rates during leak testing of Annulus hold up test were high ~ 30m³/hr, indicating that the leak was directly communicating to formation behind.

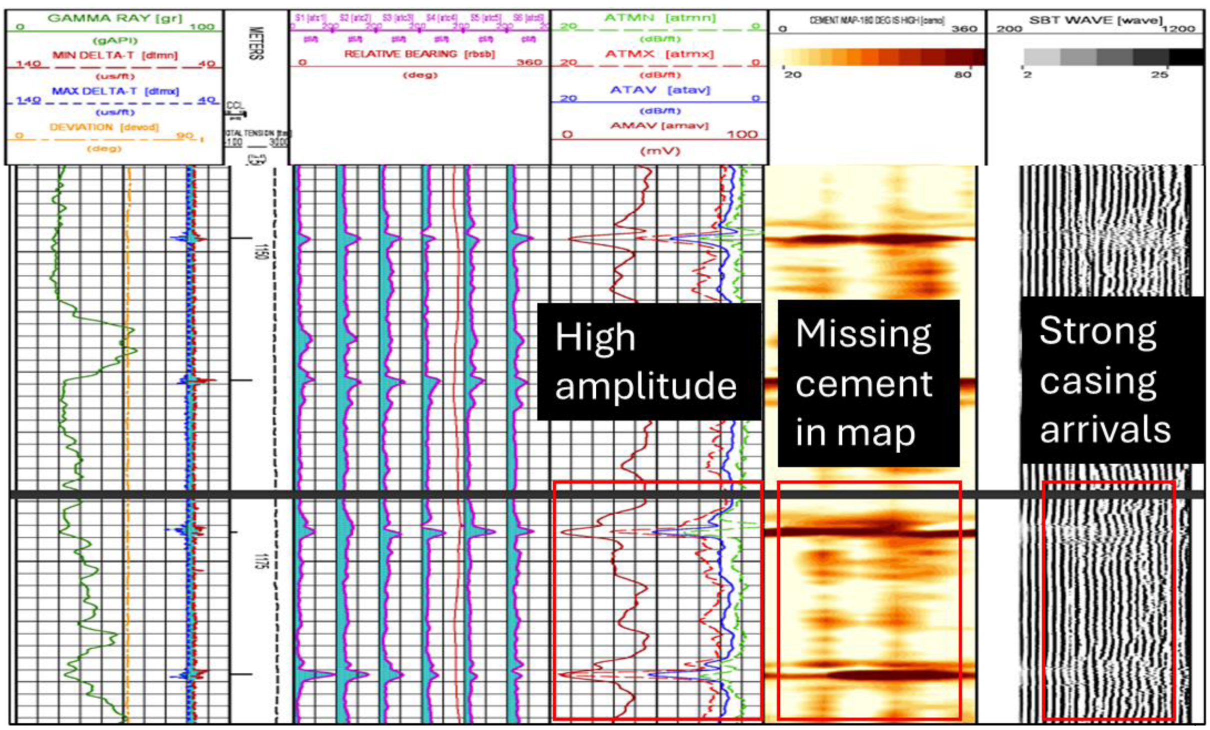

Figure 11—Unsupported section visible at leak depths

The data of leak along with deformity was passed onto the Geomechanics group and Reservoir & Geology team. At the same time Production Technology team conducted cause and effect analysis mentioned in CASE-A investigations.

RESERVOIR & GEOLOGY TEAM. The team found that the area was not abnormally depleted or over-pressured. However, the well trajectory crosses a fault plane right at the depth of deformation. The field has many faults created during and after deposition of reservoir and overlying zones. The field is mostly under a strike-slip regime. A snapshot of fault plane intersecting Well-B is shown in Figure 12.

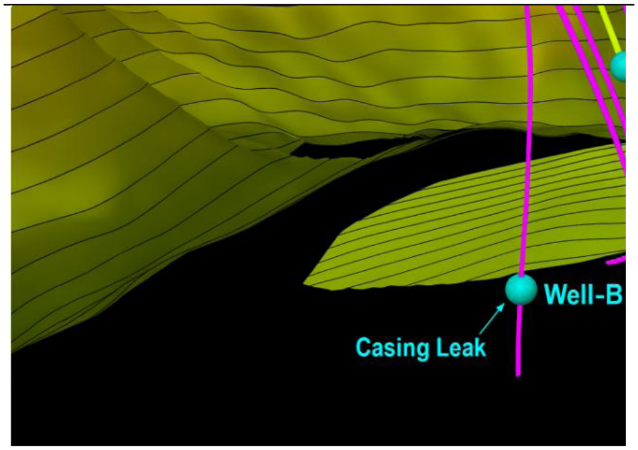

Figure 12—Fault plane-well intersection

GEOMECHANICS TEAM. The fault plane is also intersecting the nearby few wells, but no similar damage was found there. A nearby injector was also crossing the fault plane. However, the injection zone was far away from fault plane and cement quality between injection zone and fault plane intersection was exceptionally good. Hence, an out-of-zone injection into the fault was ruled out. The Geomechanics team still pursued the analysis with investigation of pore pressures required to make fault critically stressed. This is the juncture at which it is to be understood whether reservoir integrity is affecting well integrity. Eventually the impact of fault-slips due to high pore pressures were ruled out but the steps of analysis are instrumental in combining well integrity to reservoir integrity. Firstly, analytical fault re-activation analysis was carried out using Mohr-Coulomb (M-C) criteria.

Failure criteria $|\tau| < C + \mu\sigma_n$, where

τ is shear stress on fault,

C is cohesion of fault,

μ is the friction coefficient of fault,

and σ_n is normal stress on fault.

Figure 13 depicts the approach to understanding at which effective stresses, the M-C envelopes shift beyond failure criterion. Data requirement for this analysis is mentioned in table 1 below. The sources of this data in the field are also mentioned.

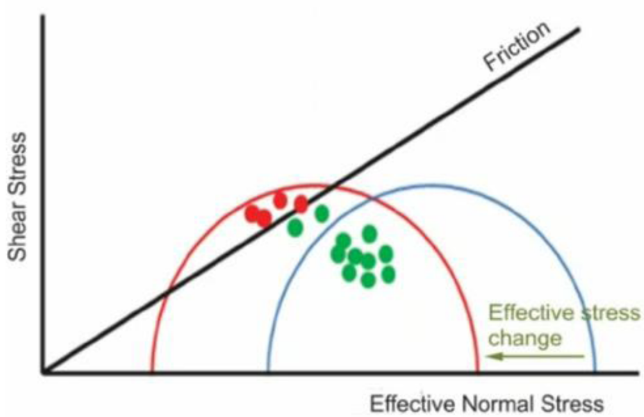

Figure 13—Depiction of MC failure criterion for fault reactivation

Variable	Data Source
Principal stresses	XLOT (eXtended Leak off test) values, DFITs and Minifrac data from hydraulic fracturing campaign
Stress Orientation	Borehole breakouts and image logs
Fault orientation	Seismic interpretation and structural maps.
Cohesion	Lab tests
Friciton angle	Lab tests
Pore-pressures	Formation tests (DST, Formation testing etc.), Sonic logs
Rock properties	Cores, Sonic logs

After the analysis it was found that pressures at which fault -reactivation might happen are excessively high beyond the operation boundary. Such pressures are not possible and were never discovered at fault plane during in-fill drilling campaign.

In absence of full field 3D geomechanical models, small scale geomechanical models were created on ANSYS software for Finite Element Analysis (FEA) modelling. An unrealistic situation of very high pore-pressure was created to mimic fault slip and understand worst case loading on Well Architecture. Failing to re-produce all the entire work here, one such case is presented here The case was run with inputs of fault angle of 30 degrees, friction coefficient of 0.6, stress magnitudes and directions defined at fault plane, at over pressure condition of 500 psig (Figure 14). The resulting von-Mises stress (Figure 15) at well level were found to be way below the strength of production casing, while limited elastic shear sliding was also observed at the simulated fault plane (Figure 16).

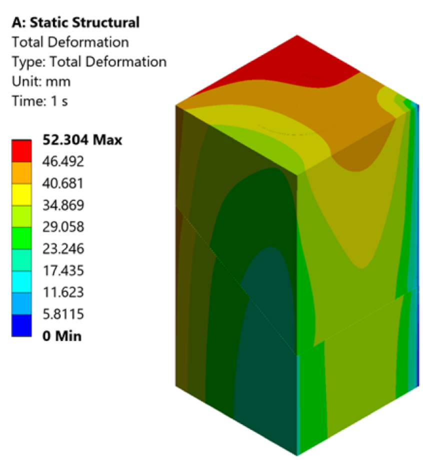

Figure 14—Small scale geomechanical model (50x50x100m) with a fault in the middle

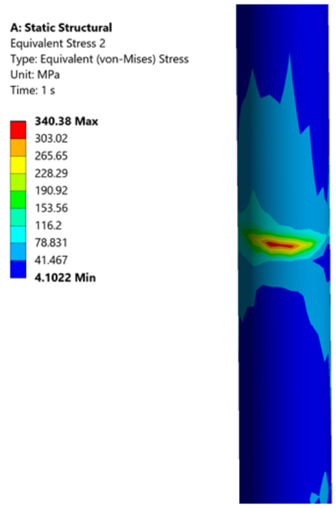

Figure 15—Von-Mises stress along case

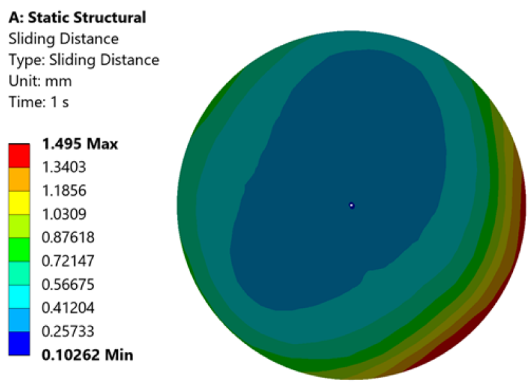

Figure 16—Sliding shear distance along the simulated fault plane

The above analysis ruled out the possibility of deformation beyond the well integrity, caused by the action of increased pressure alone. Based on these findings, it was thus decided to install shallow and deep geophone for micro-seismic monitoring and analysis in the field. The installation is in progress during the submission of this paper. A very extensive workflow for micro-seismic monitoring is mentioned by PDO reference (Wahayabi 2021) in their paper. The proposed workflows are based on similar approach and are beyond the scope of this paper. Current geomechanical findings are in line with other published before when analyzing casing deformation in tectonic setting reference (Last 2002)

WELL INTEGRITY, PRODUCTION TECHNOLOGY & WELL CONSTRUCTION GROUP. The cause of leak was not fully understood. As discussed, it was thus decided to install micro-seismics and keep monitoring wells around the fault plane. The group decided to straddle the affected zone using straddle production packers. Since the production casing above the deformed depth is healthy, it was correct to continue operating well on jet pump completion.

STEP-4 Implementation

1. Well was recompleted again on jet pump artificial lift with straddle packer set across leak point.
2. Decision to put the area under micro-seismic monitoring.
3. Fault plane pressure is closely monitored in future drilling programs.

Recommendations

The integrated workflow demonstrated in this study has proven effective in diagnosing and mitigating complex Well #And reservoir integrity issues in mature EOR fields. Based on the insights gained from Mangala field cases and supporting geomechanical analysis, the following recommendations are proposed:

1. Multi-disciplinary approach for resolving well integrity is the correct way of bringing together all teams to evaluate integrity holistically. Well Construction team, Reservoir Development, Geology and Geomechanics team should be part of steering committee to solve the issues.
2. Diagnostic tools such as caliper logs, temperature logs, corrosion logs, pressure tests, and noise logs can be used intelligently to determine the starting point for investigation. Proactive use of such tools can provide valuable insights.
3. Standardize RCA using structured tools like fishbone diagrams and multi-domain data-overlays. Also maintain a digital repository of integrity events, root causes, and corrective actions to support learning and predictive analytics.
4. Artificial lift systems (Jet Pump V/s PCP) can be re-assessed to operate well on optimized productivity but with compliance to integrity barriers.
5. Mandate the use of Well #Barrier diagrams for all wells, updated after any workover or integrity event.
6. Ensure compliance with dual-barrier policy under all operating conditions, in line with international standards.
7. Update geomechanical models regularly using field data from deformation logs, seismic surveys, and fault activation studies.
8. Conduct corrosion log runs during workovers in wells adjacent to confirmed integrity failures.
9. Install shallow and deep geophones in areas with active faulting to detect microseismic events and fault reactivation.
10. In absence of full-field 3D geomechanical models, use FEA tools like ANSYS to simulate worst-case loading scenarios and validate casing design margins.

Abbreviations

EOR – Enhanced Oil Recovery
ALS – Artificial Lift System
PCP – Progressive Cavity Pump
ESP – Electrical Submersible Pump
MEM – Mechanical Earth Model
FEA – Finite Element Analysis
ppf – pound per foot
WIMS – Well Integrity Management System
psig – pounds per square inch (pressure unit)
SAP – Sustained Annulus pressure
MINAP – Minimum Allowable Annulus Pressure
MAASP – Maximum Allowable Annulus Surface Pressure
MAAOP – Maximum Allowable Annulus Operating Pressure
Ca – Calcium element
Mg – Magnesium element
THP – Tubing Head Pressure
RCA – Root Cause Analysis
AHT – Annulus Hold-up Test
SSD – Sliding Side Door

mMD – meter, Measured Depth (well depth along well trajectory)
ID – Internal Diameter
OD – Outside Diameter
XLOT – eXtended Leak-Off Test
DFIT – Diagnostic Fracture Injection Test
DST – Drill Stem Test

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